Variables

$X = 10$

Types of Variable

Quantitative

ⓐ Descret quan. var.
(whole number)

ⓑ Continous quan. var.
(Decimal value)

qualitative

ⓐ Nominal qua. var.
(Y/N - M/F - T/F)

ⓑ Ordinal qua. var.
(10th, 12th, grad PG, PhD.)

Age	Bank a/c	Salary	Grade	Result
21.5	1	10K	B	Pass
	2	20K	A	Fail
	3	30.5K	B+	
	4			

Random variable

$$Y \approx [10, 20, 30, 40]$$

Descret Random variable

(whole, categorical)

$Y \approx [A, B, C]$

$[1, 2, 3]$

Continous Random variable

(Decimal value)

$Y \approx [3, 1.5, 2.75]$

0.05

	Age	Salary	A/c	Result	Gender
1	10	10k	1	P	M
2	20	20k	4	F	F
3	30	40k	3	P	M

~~X_1~~	X_2	X_3	X_4	X_5	Y
ID	Age	Education	I.Q.	Grade	Placement
–	–	–	–	–	–
–	–	–	–	–	–
–	–				–
		–	–	–	–
–	–	–	–	–	–
–	–	–	–	–	–
–		–	–	–	–

label data - target single

X_2, X_3, X_4, X_5 – Independent variable

Y – Dependent variable

Correlation - $X_2 - Y$ $\quad$ $X_2 - X_3$

$X_3 - Y$ $\quad$ $X_3 - X_4$

$X_4 - Y$ $\quad$ $X_4 - X_5$

$X_5 - Y$ $\quad$ $X_4 - X_2$

$X_5 - X_2$

$X_3 - X_5$

Inferential statistics

Probability :- Probability is the likelihood of the event.

$$Pr = \frac{\text{no. of way it can happen}}{\text{Total no. of outcome}}$$

Coin = H | T

$$Pr(H) = \frac{1}{2}$$

$$= 50\%$$

$$\frac{1}{2} = 0.5 = 50\%$$

Dice = 1, 2, 3, 4, 5, 6

$$Pr(5) = \frac{1}{6}$$

$$= 0.16$$

$$= 16\%$$

Type of probability

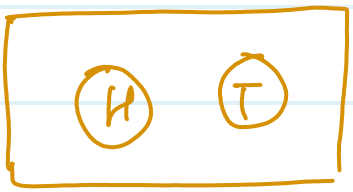
① mutually exclusive
② Non-mutually exclusive

① mutually exclusive –
(single outcome)

Coin = H/T

Dice = 1/2/3/4/5/6

T/F = 1/2

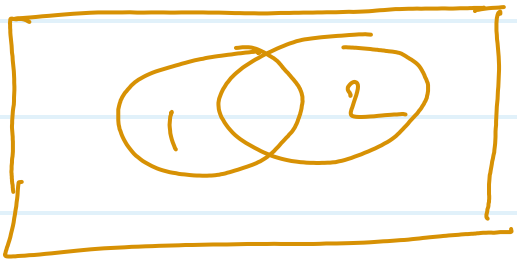


② Non-mutually exclusive
(more than one outcome)

Deck of card

♡ ♢ ♤ ♧

K, J, Q, A, 2, 3 10



$$Pr(K) = \frac{4}{52}$$

$$Pr(K \text{ and } ♢) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

Rule of probability

① Additive rule
② multiplicative rule.

① Additive rule

Type-I mutually exclusive

$$P(A \text{ or } B) = P(A) + P(B)$$

Eg:- In a Dice probability of 2 and 5

$$P(2 \text{ or } 5) = P(2) + P(5)$$

$$= \frac{1}{6} + \frac{1}{6} + P(3)$$

$$= \frac{2}{6}$$

$$= \frac{1}{3} \Rightarrow 0.33 \doteq 33\%$$

Type - II non-mutually exclusive

$$P(A+B) = P(A) + P(B) - P(A \text{ and } B)$$

eg:- calculate probability of king and club in cards.

$$P(K \text{ and } \clubsuit) = P(K) + P(\text{club}) - P(K \text{ and club})$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

$$= \frac{16}{52}$$

$$= \frac{4}{13}$$

$$= 0.307$$

$$= 30.7\ \%$$

② multiplicative rule

Type-I Independent even

$P(A)$ or $P(B)$ $= P(A) \cdot P(B)$

Eg:- Coin (H) and Dice (4)

$P(H)$ and $P(4) = \frac{1}{2} \times \frac{1}{6}$

$= \frac{1}{12}$

$= 0.08$

$= 8\%$

Type-II Dependent event

$P(A)$ or $P(B) = P(A) \cdot P(B/A)$

Eg:- $P(J)$ and $P(K)$

$P(J)$ and $P(K) = \frac{4}{52} \times \frac{4}{51}$

$$= \frac{4}{663}$$

$$= 0.006$$

$$= 0.6\%$$

A B C D

AB	BA
AC	CA
AD	DA
BC	CB
BD	DB
CD	DC

① Combination

$$^{n}C_{r} = \frac{n!}{r!(n-r)!}$$

$$n = 4$$
$$r = 2$$

$$= \frac{4!}{2!(4-2)!}$$

$$= \frac{4!}{2!\,.\,2!}$$

$$\Rightarrow \frac{\cancel{4} \times 3 \times \cancel{2 \times 1}}{\cancel{(2 \times 1)}\ \cancel{(2 \times 1)}}$$

$$\Rightarrow 6$$

$n = 8$

$r = 3$

$$\Rightarrow \frac{8!}{3!\,(8-3)!}$$

$$\Rightarrow \frac{8 \times 7 \times \cancel{6} \times \cancel{5} \times \cancel{4} \times \cancel{3!}}{\cancel{3!} \times \cancel{5} \times \cancel{4} \times \cancel{3} \times \cancel{2} \times 1}$$

$$\Rightarrow 56$$

(2) Permutation $\Rightarrow$

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

$n = 6$

$r = 3$

$$\Rightarrow \frac{6!}{(6-3)!} \Rightarrow \frac{6 \times 5 \times 4 \times \cancel{3 \times 2 \times 1}}{\cancel{3 \times 2 \times 1}}$$

$$\Rightarrow 120$$

✱ Confidence Interval

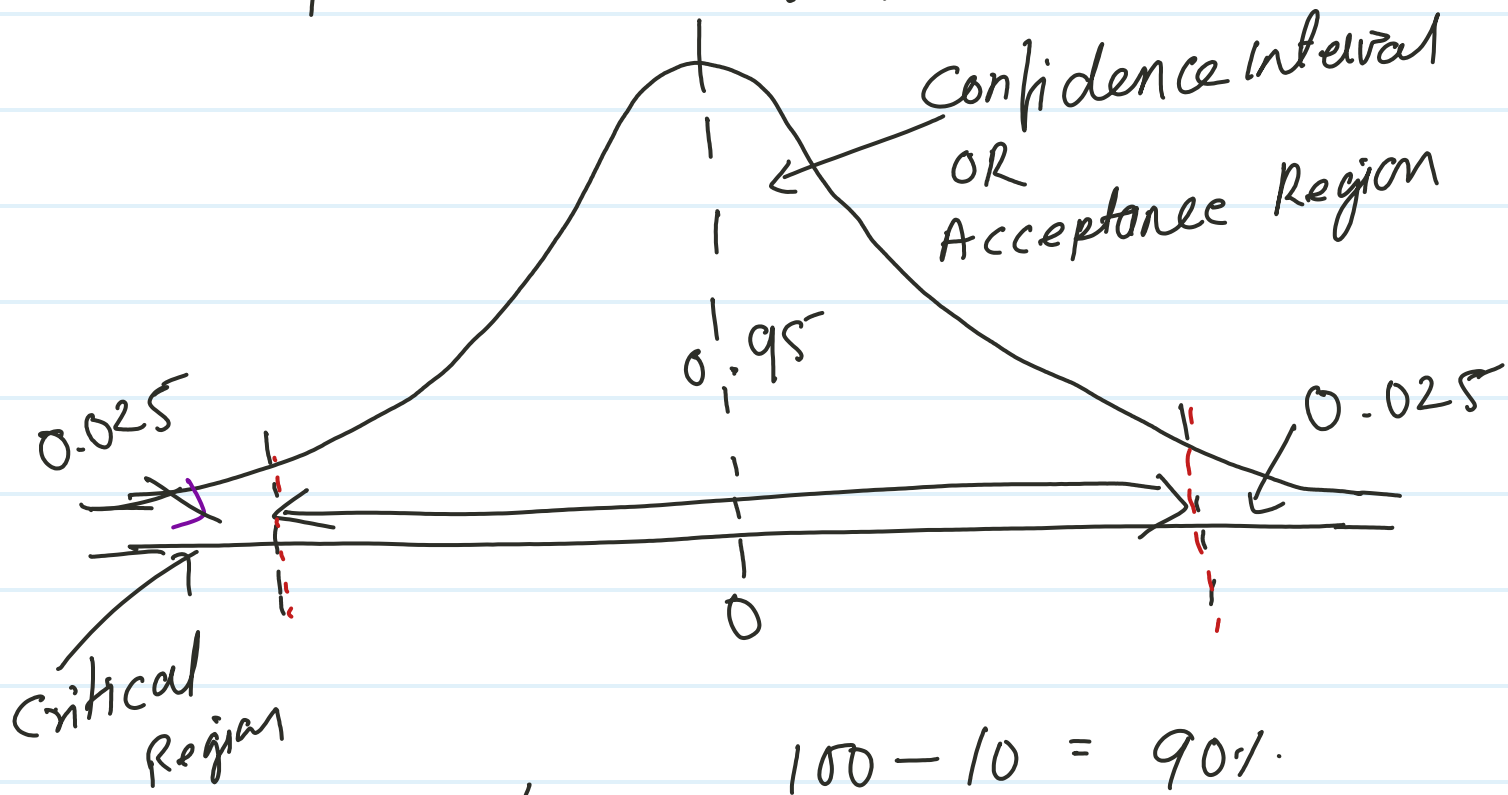
Dataset → 1000

α = Domain expert

= 5 % significant value

Critical region

100 − 5 = 95 %

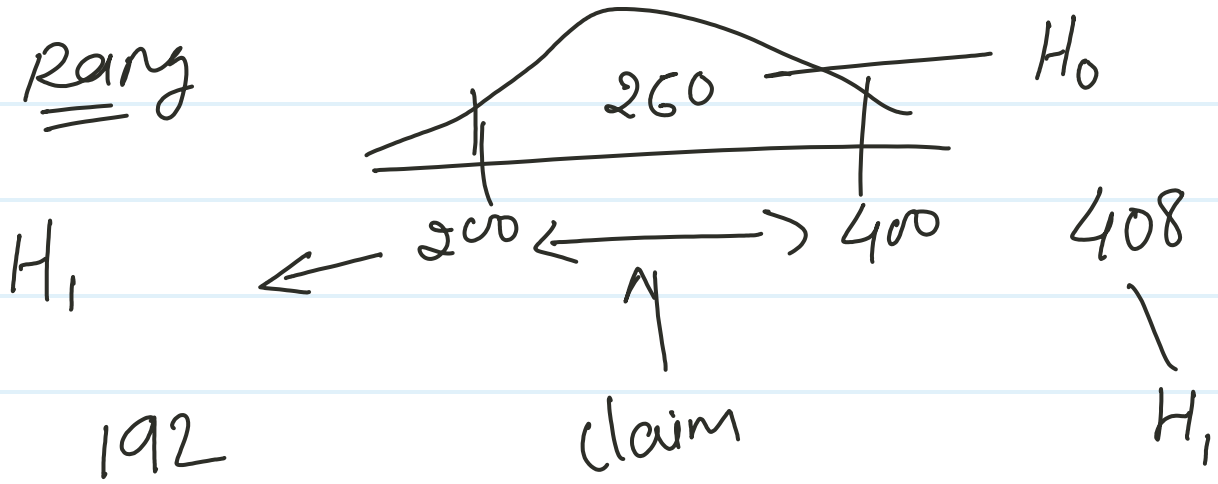


100 − 10 = 90 %

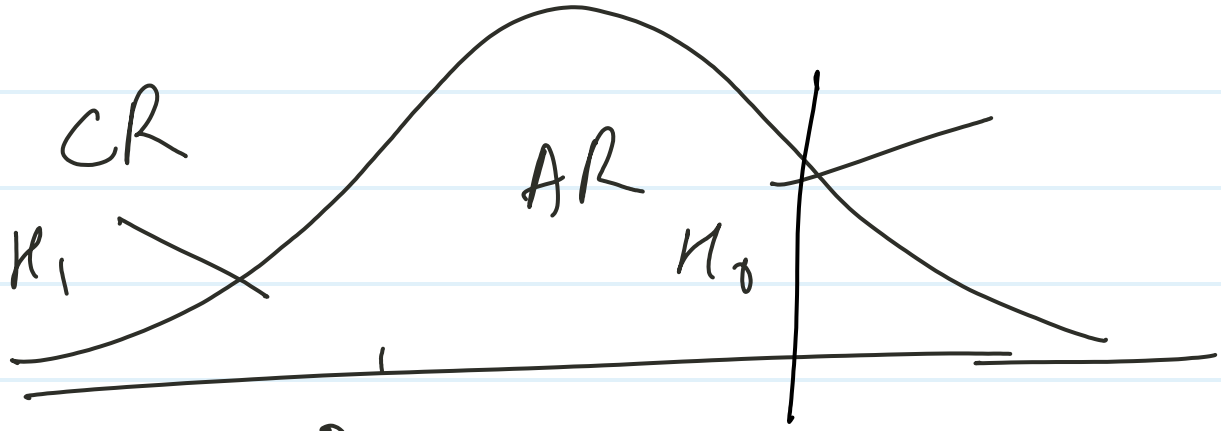
α = 10 %

$\alpha/2$

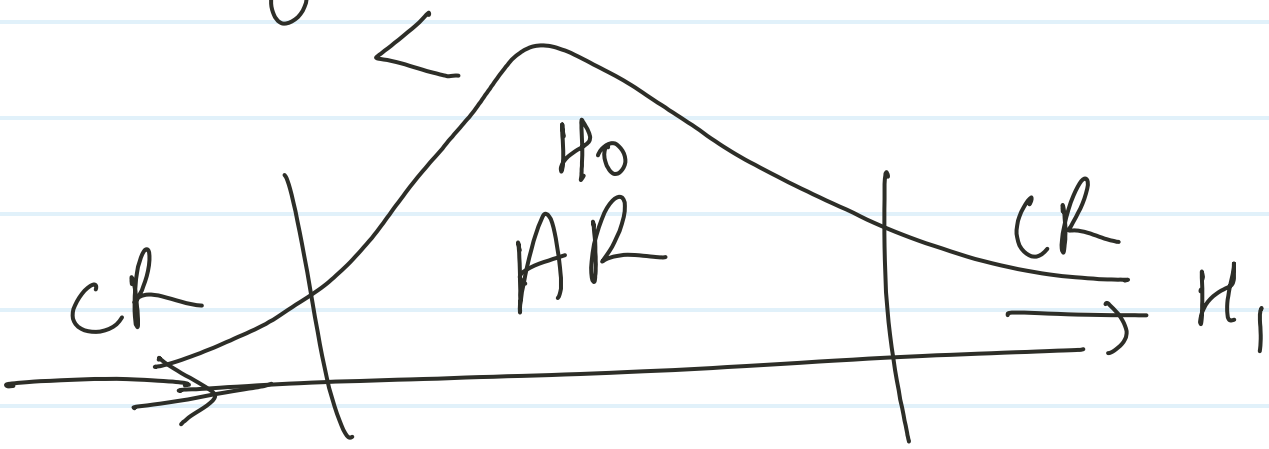
Rang



Two tail and one-ti



0



$$P = \alpha$$

$$0.06 > 0.05$$

$$H_0$$

$$0.03 < 0.05$$

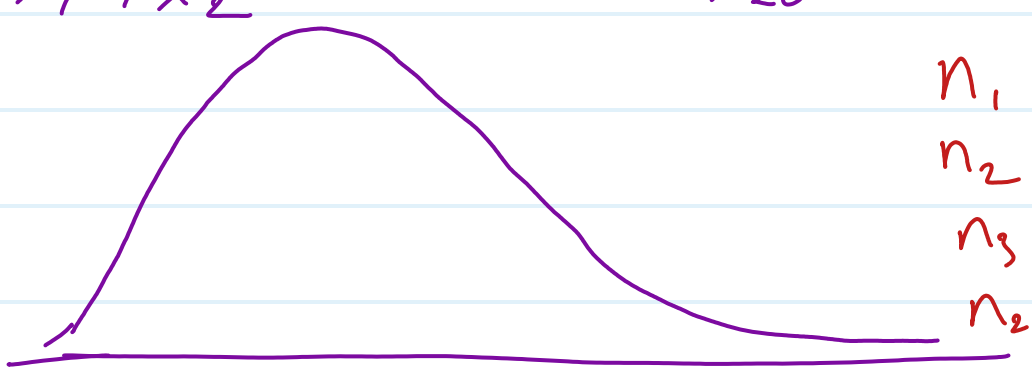
$$H_1$$

☆ Center Limit theorem

50000

$n_1, n_2 - - - - - - - n_{20}$ — n_{60}

$\bar{x}_1, \bar{x}_2 - - - - - - \bar{x}_{20}$ $\bar{x}_{60}$

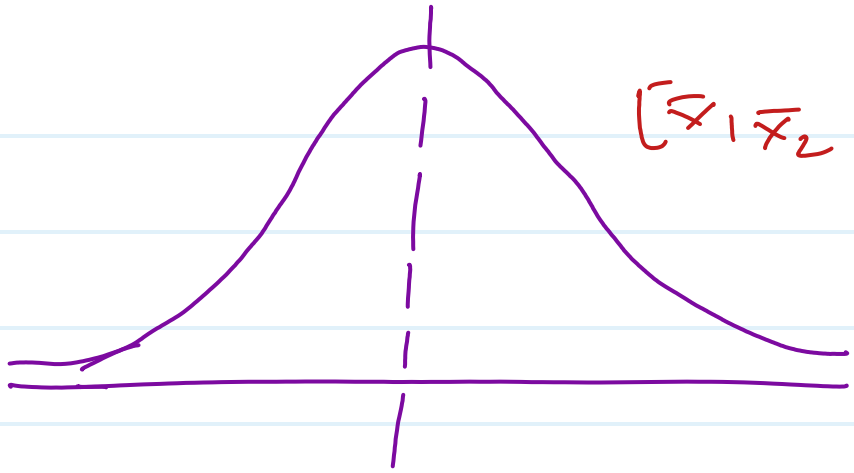


$n_1 = 30 = \bar{x}_1$

$n_2 = 30 \quad \bar{x}_2$

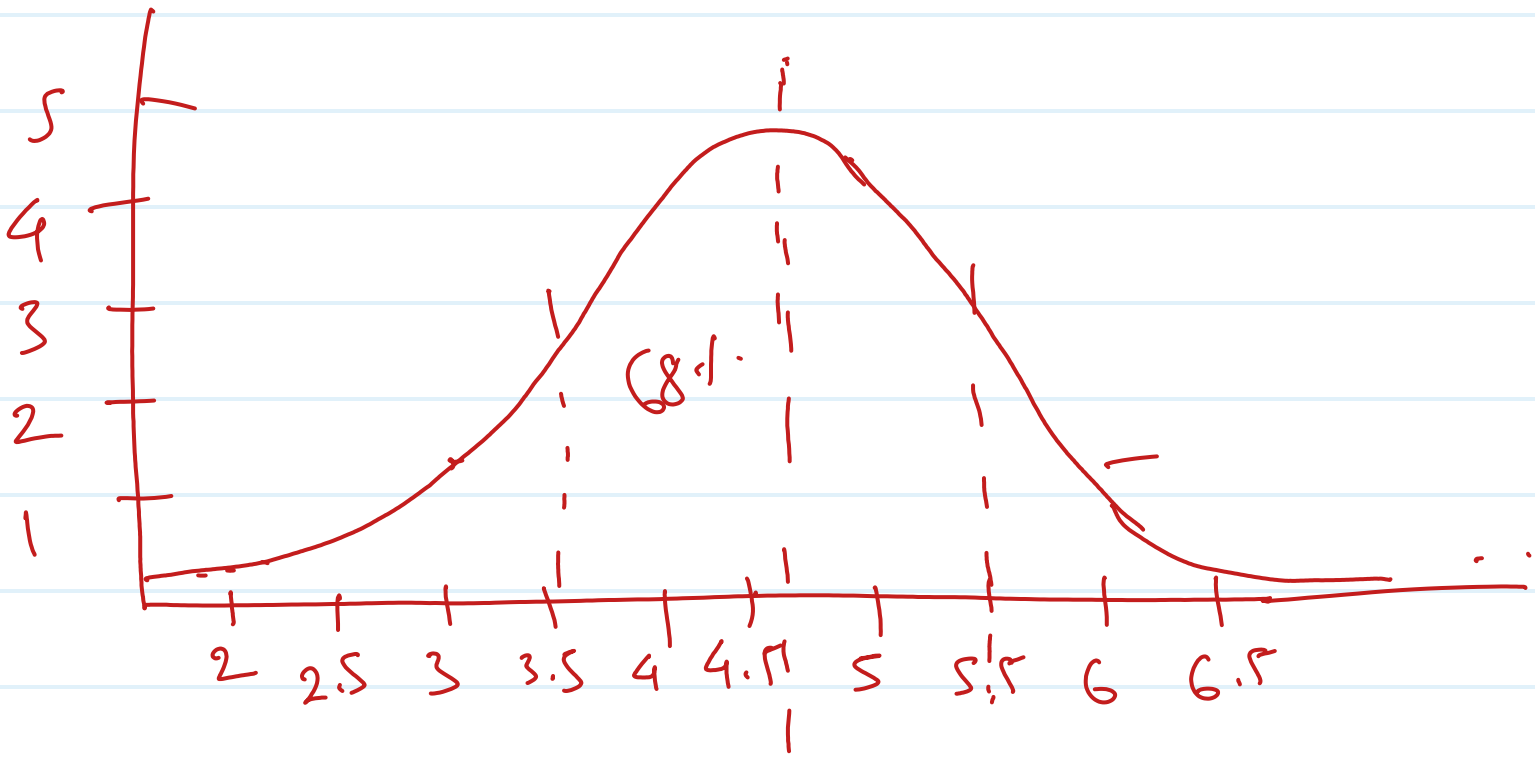
$n_3 = 30 \quad \bar{x}_3$

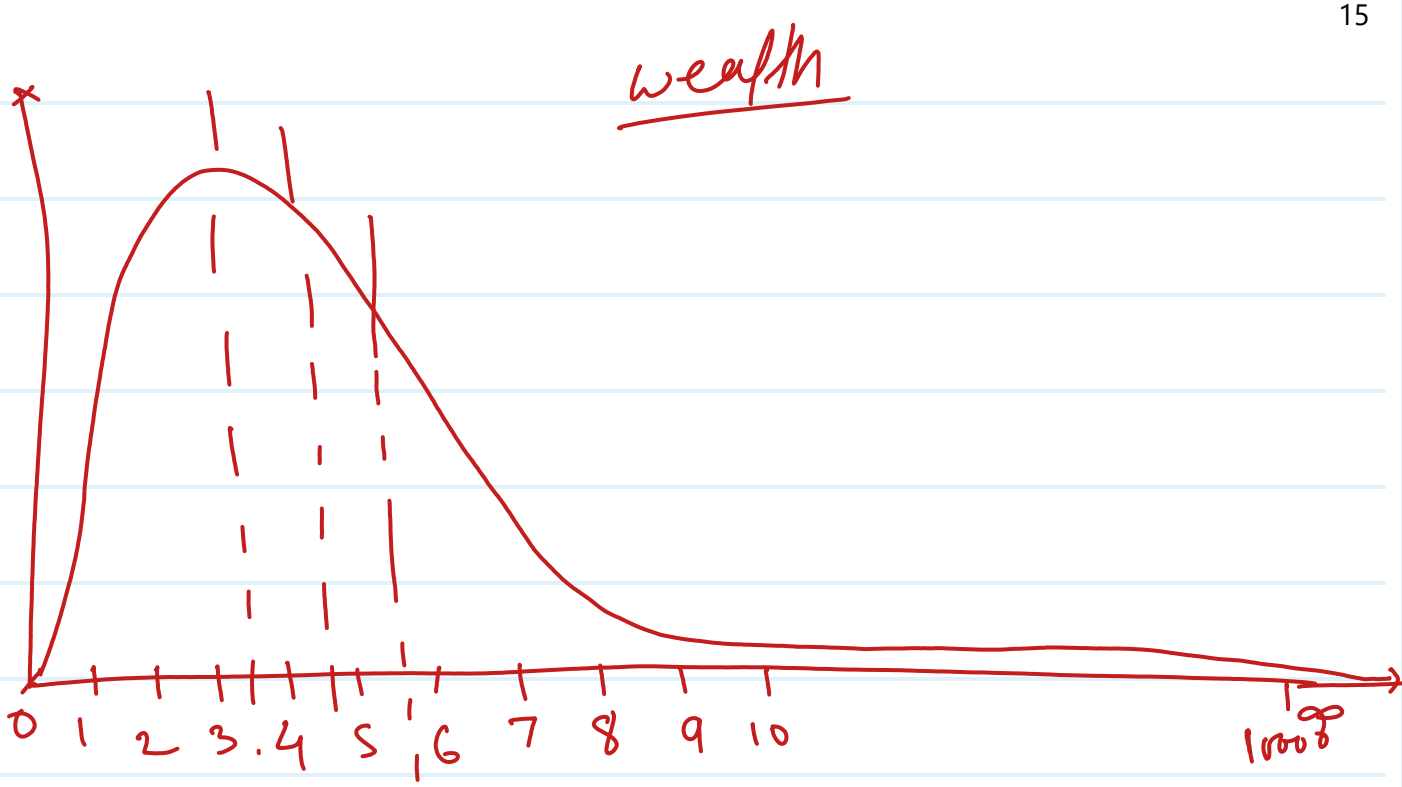
$n_{20} = 30 \quad \bar{x}_{20}$

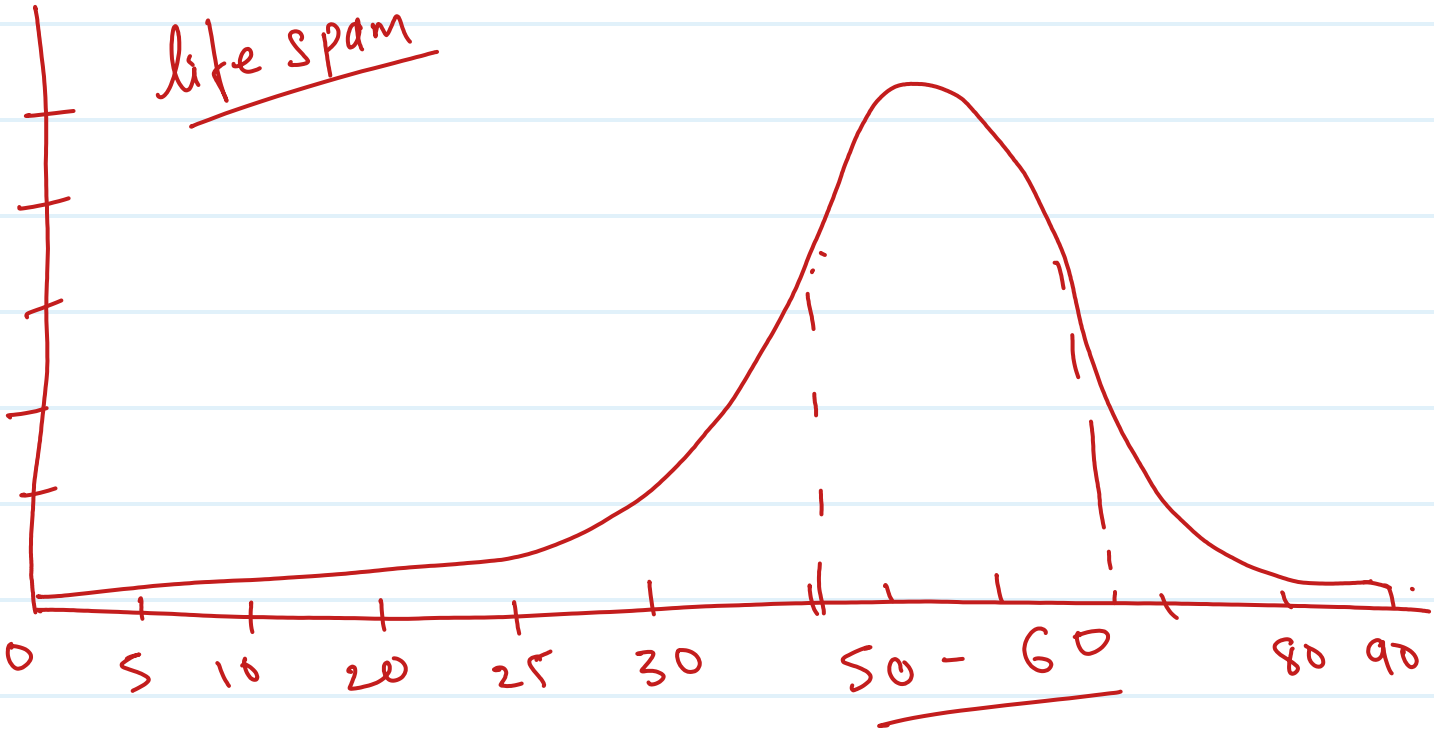


$[x_1 x_2 \ldots\ldots x_{50}]$

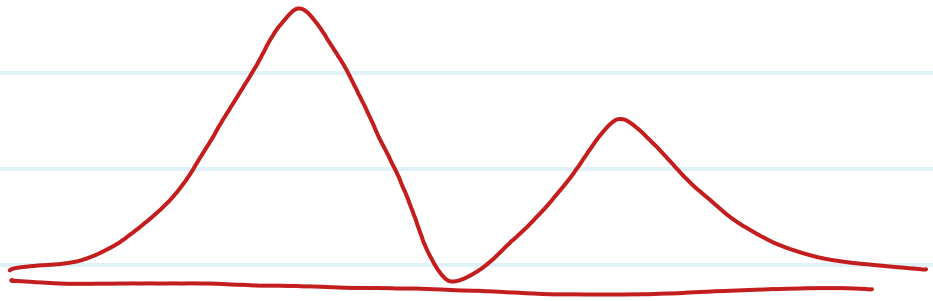
math	70	= 31.8
science	60	= 27.2
Eng	50	= 22.7
History	40	= 18.18
	220 = 100%	100




wealth
0
1
2
3
4
5
6
7
8
9
10
1000
∞


life span
0
5
10
20
25
30
50 - 60
80
90

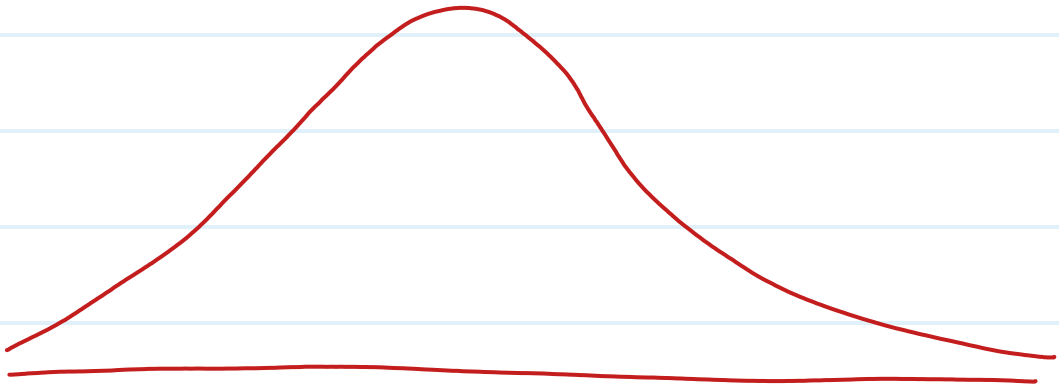
$\Rightarrow 10000$



$n_1 = 25 = \bar{x}_1$

$n_2 = 25 = \bar{x}_2$

$n_{30} = 25 = \bar{x}_{30}$

$10 - 7 - 3$

$[\bar{x}_1, \bar{x}_2, \bar{x}_3 - - - - - - - \bar{x}_{30}]$